

PHASE 2 : Project Cost Estimate — Module Specification

Objective

The Project Cost Estimate module (informally referred to as the Fund Report) is the primary reporting component of Phase 2 of the Integrated Digital Business Platform (IDBP).

It enables field-level staff (Junior Engineers) to prepare itemised cost estimates for ongoing projects, which then flow through a two-stage approval hierarchy — Zonal Office (Area Office) then Head Office — before being considered final.

Roles and Workflow

The Fund Report follows a Maker → Checker → Approver model.

Role	Title	Responsibility
Maker	Junior Engineer (JE)	Creates and submits the estimate
Checker	Area Office (AO) / Zonal Office	Reviews each line item and marks approval per row
Approver	Head Office (HO)	Final review, per-row approval, and source of purchase assignment

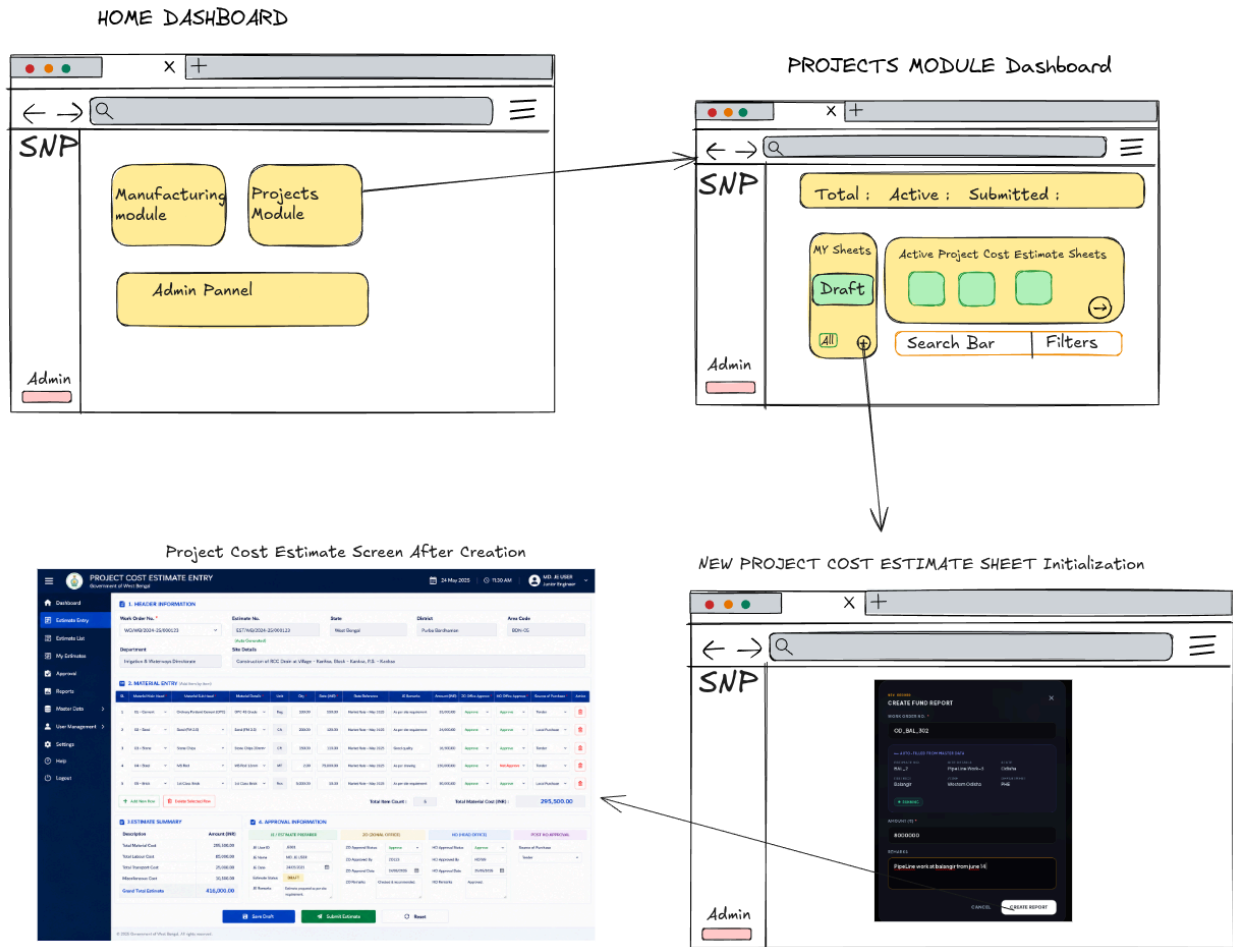
Workflow Rules

- A JE can create and submit a Fund Report but cannot edit or delete it after submission.
- The Area Office can edit or soft-delete a submitted report before Area Office approval.
- The Head Office can view a report only after Area Office submission.
- After Area Office approval, only the Head Office can edit or soft-delete the report.

- No report can be permanently deleted by any user, including Admin. Only soft deletion is permitted.

Workflow

```
JE Creates Estimate
|
JE Adds Line Items
|
JE Submits Estimate
|
A0 Reviews Each Line Item
A0 marks each row: Approve / Not Approve
A0 enters remarks if required
|
A0 Confirms and Submits to H0
|
H0 Reviews Each Line Item
H0 marks each row: Approve / Not Approve
H0 enters remarks if required
H0 selects Source of Purchase (header-level, for the whole estimate)
|
H0 Final Confirmation
```



Estimate Structure

A Project Cost Estimate consists of three parts:

1. Estimate Header — entered once, applies to the entire estimate
2. Estimate Line Items — repeating rows, one per material or cost item
3. Approval Information — system-stamped and user-entered, per stage

Each Project Cost Estimate is linked to exactly one Work Order No. A Work Order may have multiple estimates associated with it. The Work Order No. becomes immutable after the estimate is created.

Form Schema

Part 1 — Estimate Header

The following fields form the header of the estimate. They are entered or auto-populated once at the time of estimate creation and apply to the entire estimate.

Auto-Populated Fields (triggered by Work Order No. selection)

When the user selects a Work Order No., the following fields are automatically populated from the Master Data Sheet and become read-only on the form.

Field Name	Input Type	Data Source
Work_Order_No	Dropdown	Master Data
Estimate_No	Auto	Master Data
State	Auto	Master Data
District	Auto	Master Data
Area_Code	Auto	Master Data (Zone column)
Department	Auto	Master Data
Site_Details	Auto	Master Data

Note: Area_Code maps to the Zone field in the Master Data schema. The Work Order No. dropdown must only show projects with status Running or Complete Under Maintenance. Closed projects are not selectable for new estimate creation.

Initialization Fields (entered by JE at creation time)

These fields are captured in the Create Fund Report modal when the JE initialises a new estimate.

Field Name	Input Type	Notes
Area_Office_No	Text Entry	Reference number for the Area Office. Distinct from Area_Code.
JE_Remarks	Text Entry	Opening remarks or context provided by the JE at creation time

System-Stamped Header Fields (JE submission)

These are stamped automatically at the moment the JE submits the estimate. They are never editable by any user.

Field Name	Populated At	Value
JE_User_Id	On JE submission	Logged-in user's phone number
JE_Date	On JE submission	System date and time
Estimate_Status	Throughout lifecycle	System-managed — see Estimate Status section

Part 2 — Estimate Line Items

Each estimate may contain one or more line items. There is no upper limit on the number of rows; estimates may span 500 or more rows. Users can add and delete rows. Amount is always system-calculated and is never directly editable.

Cascading Dropdown Fields (from Material Data)

Material selection is a three-level cascade. Each level filters the options available in the next.

Field Name	Input Type	Depends On
Material_Main_Head	Dropdown (Level 1)	—
Material_Sub_Head	Dropdown (Level 2)	Material_Main_Head
Material_Details	Dropdown (Level 3)	Material_Sub_Head
Unit	Auto	Material_Details (from Material Data)

Manual Entry Fields

Field Name	Input Type	Notes
Qty	Number Entry	User-entered quantity
Rate	Number Entry	User-entered rate
Rate_Reference	Text Entry	Source or basis of the rate (e.g. DSR, market rate)
Amount	Auto	Calculated: Rate x Qty. Never directly editable.
Source_of_Purchase	Dropdown	Filled by JE per row. Options drawn from the Purchase_Data reference table, which is maintained by the ERP Admin.

Per-Row Approval Fields

Approval is recorded at the line-item level. Each row carries its own approval status from both the AO and HO stages. These fields are populated by the respective approving role and are not editable by any other user.

Field Name	Input Type	Populated By	Notes
ZO_Office_Approve	Dropdown	Area Office	Options: Approve / Not Approve
AO_Remarks	Text Entry	Area Office	Optional on approval; recommended on rejection
HO_Office_Approve	Dropdown	Head Office	Options: Approve / Not Approve
HO_Remarks	Text Entry	Head Office	Optional on approval; recommended on rejection

Part 3 — Approval Information

JE Stage Fields

Field Name	Input Type	Notes
JE_User_Id	Auto	Logged-in user's phone number, stamped on submission
JE_Name	Auto	Resolved from User ID
JE_Date	Auto	System date and time at submission
Estimate_Status	Auto	Reflects current workflow state

AO / Zonal Office Stage Fields

Field Name	Input Type	Notes
ZO_Approval_Status	Dropdown	Approve / Not Approve — header-level confirmation after row review
ZO_Approved_By	Auto	Logged-in user's phone number, stamped on AO confirmation
ZO_Approval_Date	Auto	System date and time at AO confirmation
ZO_Remarks	Text Entry	Overall remarks by the AO on the estimate

HO / Head Office Stage Fields

Field Name	Input Type	Notes
HO_Approval_Status	Dropdown	Approve / Not Approve — header-level confirmation after row review
HO_Approved_By	Auto	Logged-in user's phone number, stamped on HO confirmation
HO_Approval_Date	Auto	System date and time at HO confirmation
HO_Remarks	Text Entry	Overall remarks by the HO on the estimate

Estimate Status

The Estimate_Status field is system-managed and is never directly editable by any user. It transitions automatically based on user actions.

Status	Triggered By
Draft	JE saves the estimate without submitting
Submitted	JE clicks Submit
Under AO Review	AO opens the estimate for review
AO Approved	AO confirms approval at header level
Rejected by AO	AO selects Not Approve at header level
Under HO Review	HO opens the estimate for review
Final Approved	HO confirms approval at header level
Rejected by HO	HO selects Not Approve at header level

Estimate_Status is visible to all roles and serves as the primary filter on the dashboard and Estimate List views.

Material Data Reference

The Material Data sheet defines the complete catalogue of items available for selection in a Fund Report. The three-level hierarchy is:

Material_Main_Head → Material_Sub_Head → Material_Details

Material_Main_Head	Material_Sub_Head(s)
Materials	Raw Materials

Material_Main_Head	Material_Sub_Head(s)
Electrical	Electrical Consumables
Mechanical	Mechanical Consumables, Mechanical Project Items
Pipes	Pipe HDPE, Pipes Steel, Pipe DI, Pipe PVC
Solar System	Solar Pump Set Material
Labour	Labour Category
Transport	Transport / Logistics
Miscellaneous	Miscellaneous Cost

Each Material_Details entry has a corresponding Unit (e.g. Bag, Mtr, Nos, Kg, Day, Trip, Cum) that is auto-filled when the item is selected.

Estimate Summary Calculation

The system automatically calculates estimate totals from all line items. All summary values are read-only and system-generated.

Summary Field	Calculation
Total Material Cost	Sum of all line items where Material_Main_Head is: Materials, Electrical, Mechanical, Pipes, or Solar System
Total Labour Cost	Sum of all line items where Material_Main_Head is Labour
Total Transport Cost	Sum of all line items where Material_Main_Head is Transport
Total Miscellaneous Cost	Sum of all line items where Material_Main_Head is Miscellaneous
Grand Total Estimate	Sum of all line item amounts across all categories

Note: Electrical, Mechanical, Pipes, and Solar System line items are grouped under Total Material Cost. They do not have separate summary lines.

Access Control by Project Status

Fund Reports are governed by the project's Status in the Master Data Sheet.

Project Status	Create Estimates	Edit Estimates	View Estimates
Running	Yes	Yes	Yes

Project Status	Create Estimates	Edit Estimates	View Estimates
Closed	No	No	Yes
Complete Under Maintenance	Yes	Yes	Yes

A report referencing a Closed project automatically becomes read-only for all users until the Admin changes the project status back to Running or Complete Under Maintenance.

Permissions Matrix

Action	Junior Engineer (JE)	Area Office (AO)	Head Office (HO)
Create Fund Report	Yes	Yes	Yes
Edit Report (before JE submission)	Yes	No	No
Edit Report (after JE submission, before AO approval)	No	Yes	No
Edit Report (after AO approval)	No	No	Yes
Soft-delete Report (after JE submission, before AO approval)	No	Yes	No
Soft-delete Report (after AO approval)	No	No	Yes
View Report at all stages	Yes	Yes	Yes
Mark row-level approval (AO stage)	No	Yes	No
Mark row-level approval (HO stage)	No	No	Yes
Confirm header-level approval (AO stage)	No	Yes	No
Confirm header-level approval (HO stage)	No	No	Yes
Set Source of Purchase (per row)	No	No	Yes

Soft-delete means the record is flagged as deleted in the database and removed from standard views. It is never physically removed. Soft-deleted records remain visible in audit trail views. No user, including Admin, can permanently delete a Fund Report record.

Logical Data Model

Table: project_cost_estimates

One row per estimate.

Column	Type	Notes
estimate_id	UUID / PK	
work_order_no	FK to Master Data	Immutable after creation
area_office_no	String	Entered at initialization
estimate_status	Enum	See Estimate Status section
je_user_id	String	Stamped at submission
je_date	Timestamp	Stamped at submission
je_remarks	Text	
zo_approval_status	Enum (Approve / Not Approve / Null)	
zo_approved_by	String	
zo_approval_date	Timestamp	
zo_remarks	Text	
ho_approval_status	Enum (Approve / Not Approve / Null)	
ho_approved_by	String	
ho_approval_date	Timestamp	
ho_remarks	Text	
is_deleted	Boolean	Soft-delete flag
created_at	Timestamp	
updated_at	Timestamp	

Table: project_cost_estimate_items

One row per material line item.

Column	Type	Notes
item_id	UUID / PK	
estimate_id	FK to project_cost_estimates	
material_main_head	String	
material_sub_head	String	
material_details	String	
unit	String	Auto from Material Data
qty	Decimal	
rate	Decimal	
rate_reference	String	
amount	Decimal	Computed: rate x qty
source_of_purchase	FK to Purchase_Data	Set by HO per row. References the Purchase_Data table maintained by ERP Admin.
zo_office_approve	Enum (Approve / Not Approve / Null)	Row-level, set by AO
ao_remarks	Text	
ho_office_approve	Enum (Approve / Not Approve / Null)	Row-level, set by HO
ho_remarks	Text	
is_deleted	Boolean	Soft-delete flag

Data Integrity Rules

- A Fund Report record cannot be permanently deleted by any user. Only soft deletion is permitted.
- Work_Order_No on a submitted report is immutable and cannot be changed after the estimate is created.

- Amount is always system-calculated as Rate x Qty and is never directly editable.
 - Unit is always auto-populated from the Material Data catalogue and is never directly editable.
 - All approval timestamps and user IDs are system-stamped and cannot be manually entered or overridden.
 - Estimate_Status is system-managed and cannot be directly edited by any user.
 - Source_of_Purchase is entered by the JE at the line-item level. Its options are drawn from the Purchase_Data reference table, which is maintained exclusively by the ERP Admin.
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Future Enhancements

- Report export to PDF or Excel
 - Role-specific dashboards showing pending approvals
 - Notification and alert system for approval stage transitions
 - Historical audit trail view per Fund Report
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Clarifications Needed

Can an estimate contain a mixture of approved and rejected rows?

What does header-level approval mean if row-level approvals already exist?

Should rejected rows contribute to the Grand Total?

Can HO override an AO rejection?

Can AO modify JE-entered quantities/rates or only approve/reject them?